

A machine learning-powered browser extension that detects and alerts users to manipulative UI design (dark patterns) in real time using a soft-voting ensemble of Random Forest, Gradient Boosting, and Logistic Regression over a 2 000-sample text dataset.

What Are Dark Patterns?

Dark patterns are deceptive UI/UX techniques that exploit cognitive biases to manipulate users into unintended actions — purchases, subscriptions, or data sharing — without explicit, informed consent. Common examples include urgency messages ("Only 1 item left!"), forced continuity (auto-renewals), confirmshaming, and hidden costs revealed at checkout.

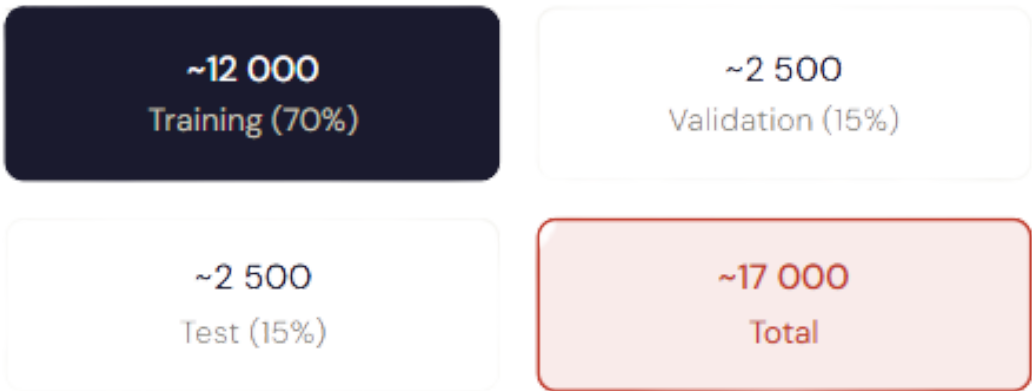
Where the Data Comes From

No standard dark-pattern corpus existed, so a custom dataset was assembled from 7 sources using web crawling, manual annotation, and AI-assisted generation.

Source	Type	Size	DP Types
Li et al. (2024) DPAF	Image+Text	5 561	68
Mathur et al. (2019)	Crawled web	53 000	15
AidUI (Mansur 2023)	Screenshots	501	10
UIGuard (Chen 2023)	Mobile UI	6 352	14
Nouwens et al. (2020)	HTML/text	10 000	Cookie
Kaur et al. (2025)	Shopping text	N/D	Shopping
Lin et al. (2026) GPT	Multimodal	Cross	65

Training / Validation / Test Split

Stratified split ensures every dark pattern class is represented in each fold. Test set is fully held-out never used for hyperparameter tuning.



Primary Architecture

Multimodal BERT + CNN Hybrid



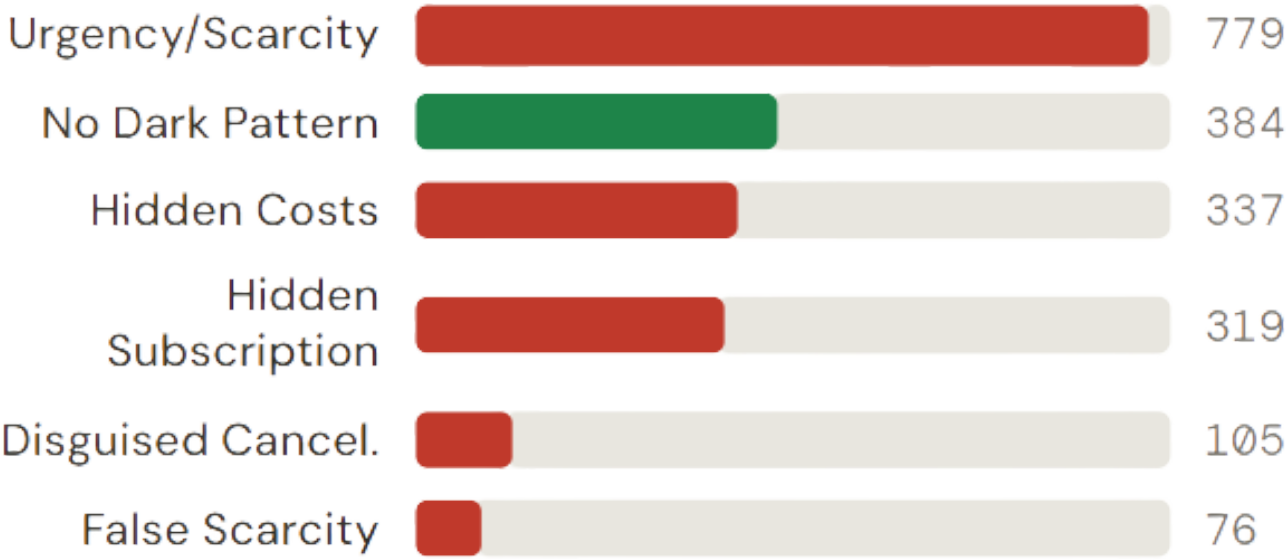
BERT captures semantic nuance beyond keyword matching — achieving 97% accuracy in Kaur et al. (2025) vs. 95.4% for RF and 95.8% for SVM. CNN detects visual/layout patterns invisible to text models (F1 > 0.82, AidUI)

Dataset Used

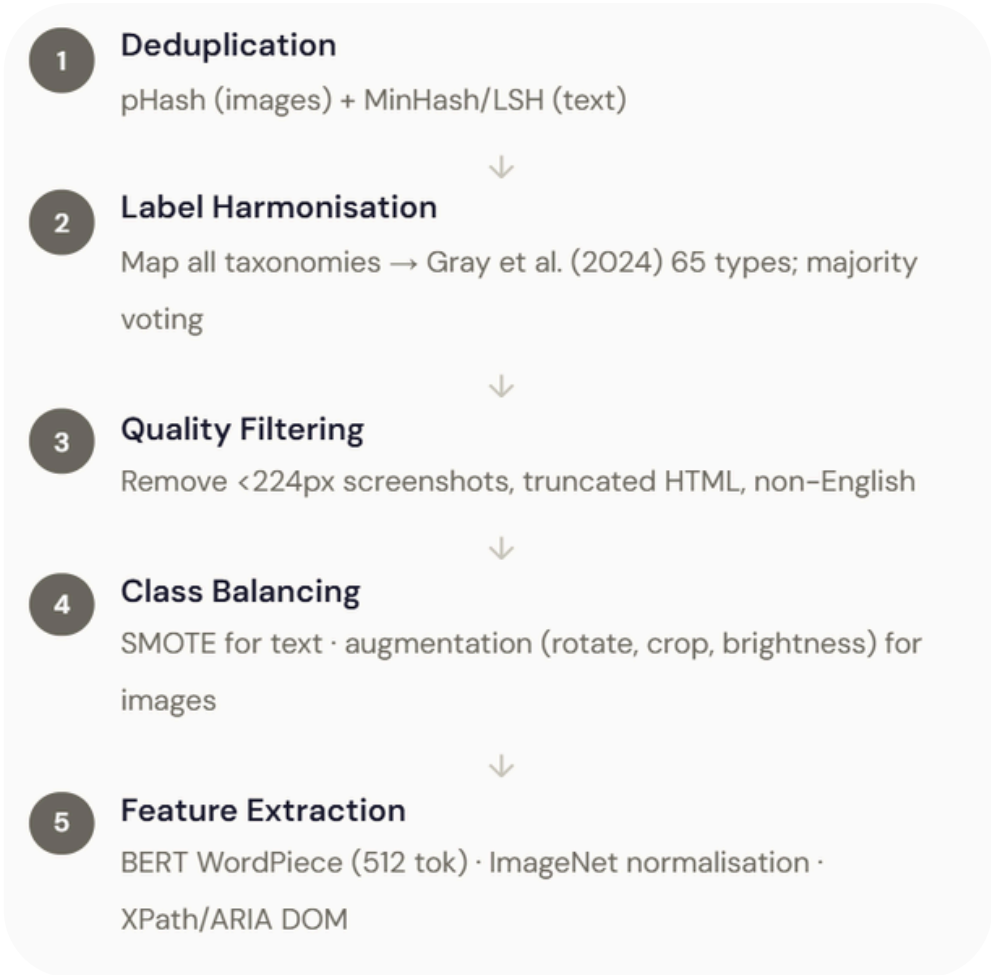
text_raw_dataset_2000.xlsx

2 000 rows · 3 columns: Website_Name, Website_Type, Description. Zero missing values. Zero duplicates after cleaning. Five website categories classified via keyword-based rule engine
Link: https://docs.google.com/spreadsheets/d/1yjVwYEl-DuraBQOoJ65oCP8l42GscMOoe5t_Je8Zrew/edit?usp=sharing

Dark Pattern Distribution



Data Cleaning Pipeline



This pipeline prepares data for a machine learning model by first removing duplicates using hashing techniques for both images and text. It then standardizes labels, filters out low-quality or irrelevant data, and balances the dataset using augmentation and SMOTE. Finally, it extracts meaningful features using techniques like BERT for text and normalized image processing for model training.

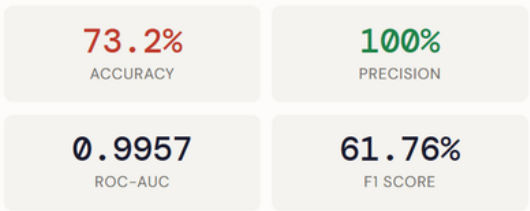
Performance Comparison

Soft-Voting Calibrated Ensemble

Combines Random Forest (weight 3) + Gradient Boosting (weight 2) + Logistic Regression (weight 1) using soft-voting probability averaging. Acts as a consensus protocol — flags only when all three models agree.

Algorithm	Accuracy	Precision	Recall	F1
Logistic Regression	68.5%	78.0%	38.0%	51.1%
Random Forest	71.2%	88.0%	52.0%	65.3%
Gradient Boosting	72.5%	96.0%	42.0%	58.4%
★ Ensemble (Ours)	73.2%	100%	44.7%	61.8%

Ensemble achieves 100% Precision (zero false alarms) and highest overall accuracy. F1 is deliberately lower than Random Forest due to the precision-recall trade-off — the right choice for a user-facing extension where false positives destroy trust.



KEY FINDING

80.8% of 2 000 websites analysed contain at least one dark pattern

E-commerce leads at 85.9%. Urgency/Scarcity (779 instances, 39%) is the single most prevalent manipulation type, far outpacing Hidden Costs (16.9%) and Hidden Subscriptions (16%).

MODEL DESIGN DECISION

100% Precision over maximum F1 - the right trade-off

In a production browser extension, a single false positive (flagging a legitimate site) would erode user trust permanently. The ensemble's strict consensus protocol eliminates false alarms entirely while maintaining competitive recall.

RESEARCH GAP ADDRESSED

5+

Known research gaps directly addressed: real-time browser detection, visual + text fusion, dynamic JS-rendered patterns, multi-domain coverage, and sub-200ms inference via ONNX quantisation.

OUTCOME

CO1 — Machine learning methodologies for cyber security

Demonstrates how supervised ML classification can be deployed as a practical, real-time cybersecurity tool to protect users from manipulative web practices — bridging academic research and browser-native protection.